

Cover letter — *American Economic Journal: Macroeconomics*

Editor

American Economic Journal: Macroeconomics

Dear Editor,

We are pleased to submit “Volcanic Forcing and the Pre-Industrial Malthusian Trap: Pathway-Heterogeneous Demographic Response and the Long Shadow of Seasonality” for consideration at the *American Economic Journal: Macroeconomics*.

The paper estimates how pre-industrial economies absorbed climate shocks along demographic and agricultural-land margins, using volcanic eruptions as exogenous forcing and a new monthly paleo-economic panel that we assemble from the ModE-RA paleo-reanalysis, HYDE 3.5 land-use grids, the eVolv2k ice-core volcanic sulfur record, the Brecke conflict catalogue, and the Allen-Nuffield wage archive (196 countries, 3,187 sub-national units, 1421–2008 CE). The headline finding is a sharply pathway-heterogeneous Malthusian demographic margin: crop-dominant late and pastoral/mixed late countries lose population at -9.7 and $-5.5 \cdot 10^{-5}$ per Tg of stratospheric sulfur injection respectively, with the magnitudes implying $\sim 1\%$ less annualised population growth in a typical Tambora-decade exposure and $\sim 10\%$ cumulative loss for a country exposed to a Tambora-class shock without intervening recovery. Both coefficients are Bonferroni-significant across twelve pathway-by-equation tests, survive re-clustering pathways on climate primitives orthogonal to HYDE outcomes, survive a Webb wild-cluster bootstrap, and survive restriction to the post-1700 panel where HYDE’s back-projection is most reliable.

The macroeconomic substance the paper delivers, in our reading, sits in three findings of direct relevance to this journal.

First, the Malthusian channel has a quantifiable welfare arithmetic. Combining the demographic-margin coefficient with the Allen real-wage Malthus coefficient ($\hat{\beta}_{\ln P} = -0.325$ on the city-buffer specification, 1421–1850), a 30 Tg Tambora-class decade implies a 3% cumulative population decline in crop-dominant late countries and a corresponding $+0.95\%$ real-wage rise for the survivors—the canonical Malthusian welfare paradox at climate-shock magnitudes. A calibrated three-equation structural model traces the cumulative impact of the 1500–1890 VSSI trajectory: in the no-volcanism counterfactual, crop-dominant late population grows by an additional $+212\%$ over the four centuries (with the surviving per-capita welfare loss of -0.32 log-units, the Malthusian-paradox demographic dividend the literature predicts).

Second, the pre-industrial climate dimension that selected which agricultural system a society adopted leaves measurable traces in modern demographic trajectories: pre-industrial inter-annual climate volatility 1421–1750 is the single strongest cross-country predictor of modern (1955–1960 to 2015–2020) population growth at $R^2 = 0.45$, surviving an extended deep-determinants battery (Neolithic distance, migratory distance, ruggedness, land area, log GDP per capita) and an absolute-latitude control, and failing a placebo when the same volatility measure is computed over the modern 1950–2008 reanalysis period. The macroeconomic implication is that pre-industrial climate regimes shape modern demographic ceilings through institutional and technological

channels we cannot disentangle in a cross-country panel, but the geographic component of the channel is identified within-country.

Third, we test the Boserupian intensification hypothesis at cross-country pre-industrial scale and report a measurement-driven null on HYDE land-use data plus a small but statistically robust signal on the independent KK10 reconstruction. The HYDE cropland-share specification is artifactually positive (driven by colonial-transition encoding in small-island countries); the cropland-area specification, while statistically robust, is fully mediated by the demographic margin (HYDE uses population as an input). On KK10, which is population-independent by construction, the high-density intensive pathway shows a Boserupian coefficient of $+4.3 \cdot 10^{-5}$ per Tg ($p = 0.047$, surviving the demographic-margin control)—the pathway where the demographic response is weakest also shows the strongest intensification response, qualitatively consistent with channel substitution. The macroeconomic implication is that the Boserupian channel exists at pre-industrial scale, but its magnitude is two orders below what the HYDE-cropland-share specifications in the existing climate-economics literature had suggested.

The methodological documentation of HYDE's two-axis identification problem (share-specification denominator instability + area-specification demographic mediation) is itself a contribution to the empirical climate-economics literature this journal has shaped. The KK10 cross-validation strategy generalises directly.

We see the paper as a contribution to three lines of work *AEJ:Macro* has been central to: (i) the macroeconomics of climate shocks at pre-industrial as well as modern scale (Burke, Hsiang and Miguel 2015; Dell, Jones and Olken 2012); (ii) the long-run consequences of agricultural pathways and climate endowments for modern development (Ashraf and Galor 2011; Nunn and Qian 2011); and (iii) the structural dynamics of the Malthusian regime and its eventual escape, including the demographic transition signature we recover in the within-country regressions.

The paper is coauthored by two researchers at the Centro de Investigación Económica, ITAM, and Universidade de Vigo / RGEA, Spain. We declare no conflicts of interest. The manuscript has not been previously published and is not under consideration elsewhere. A replication package with the ModE-RA aggregation code, the KK10 country aggregation script, the country and sub-national parquet panels, and all analysis scripts will be deposited on OSF upon acceptance.

Sincerely,

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